Beyond "Not": How Traditional and Pretrained Models Handle Implicit Negation in Sentiment Classification

Anonymous ACL submission

Abstract

Negation is a well-known challenge in sentiment analysis, yet most prior work focuses on explicit negation markers such as *not* and *never*. Implicit negation, where negative meaning is conveyed without any surface-level cue (e.g., *"I walked out after twenty minutes"*), remains largely understudied. In this paper, we investigate whether traditional models can be brought closer to pretrained model performance on implicit negation through targeted lexicon augmentation. We compare Naive Bayes and Logistic Regression (with and without an implicit negation lexicon) against fine-tuned BERT on the IMDb dataset. We stratify the test set into three subsets — normal, explicit negation, and implicit negation — and combine a performance gap analysis with a lexicon intervention experiment. Our results show that BERT alone avoids degradation on implicit negation while all traditional models degrade relative to explicit negation, and that lexicon augmentation selectively recovers traditional model performance — most effectively for Logistic Regression — but cannot fully close the gap to BERT, suggesting that contextual pretraining captures distributional patterns beyond explicit vocabulary coverage.

1 Introduction

Sentiment classification is a foundational task in natural language processing (NLP), with applications ranging from product review analysis to social media monitoring. Despite steady progress, linguistic phenomena such as negation continue to pose significant challenges. Negation can alter or entirely reverse the polarity of a sentence, causing classifiers trained on surface-level lexical patterns to fail in systematic ways.

Prior work on negation in sentiment analysis has concentrated almost exclusively on *explicit* negation — sentences containing overt negation markers such as *not*, *no*, *never*, or morphological affixes such as *un-* and *dis-*. While these cases are important, they represent only a subset of the ways in which negative sentiment can be expressed. *Implicit* negation — where negative meaning is conveyed through semantics rather than syntax — presents a harder and less studied challenge. Consider the following contrasting examples:

- **Explicit:** "This film is *not* good."
- **Implicit:** "I walked out after twenty minutes."

Both sentences convey negative sentiment, but the second contains no negation marker. A bag-of-words classifier has no mechanism to identify *walked out* as a negative signal unless it has encountered that phrasing during training with sufficient frequency. Even a pretrained contextual model may struggle if the implicit cue is rare or domain-specific.

In this paper, we address the following research question:

RQ How large is the performance gap between traditional models (Naive Bayes, Logistic Regression) and a pretrained model (BERT) on implicit negation, and does targeted lexicon augmentation narrow this gap for traditional models?

We hypothesize that (H1) all models show greater performance degradation on implicit negation than on explicit negation relative to normal sentences, with traditional models degrading more severely; and (H2) lexicon augmentation partially recovers traditional model performance on implicit cases but cannot fully close the gap to BERT, because contextual pretraining captures distributional patterns that a finite, manually constructed lexicon cannot replicate.

2 Related Work

2.1 Traditional Approaches to Sentiment Classification

The foundational work on sentiment classification using machine learning was established by Pang et al. (2002), who compared Naive Bayes, maximum entropy, and support vector machines on movie reviews, finding that these methods definitively outperform human-produced baselines while noting that sentiment classification is harder than topic-based categorization. Their bag-of-words setup remains the standard traditional baseline. Wang and Manning (2012) further showed that Naive Bayes with bigrams achieves surprisingly competitive performance, establishing it as a strong simple baseline. Both works, however, treat negation implicitly — relying on the classifier to learn negation-related patterns from training data — rather than modeling it explicitly.

2.2 The IMDb Dataset and Compositional Sentiment

We use the IMDb Large Movie Review Dataset (Maas et al., 2011), a standard benchmark for binary sentiment classification containing 50,000 reviews. Socher et al. (2013) introduced the Stanford Sentiment Treebank (SST-2) and explicitly examined the effect of negation on model performance, showing their Recursive Neural Tensor Network substantially outperformed bag-of-words baselines on negated sentences. Critically, however, their analysis was limited to syntactically explicit cases. We revisit this gap in the context of IMDb, extending the analysis to the implicit negation setting.

2.3 Negation Handling: From Rules to Pretrained Models

Negation handling has been a persistent challenge across multiple generations of NLP methods. Ilmawan et al. (2024) provide a comprehensive survey of negation handling approaches in sentiment analysis, organizing them into rule-based, machine learning, and hybrid methods. Their key finding is directly relevant to our work: there is a significant need for more studies addressing implicit negation cue detection, even within state-of-the-art BERT-based approaches. Khandelwal and Sawant (2020) proposed NegBERT, which applies transfer learning to negation cue detection and scope resolution, achieving state-of-the-art F1 scores across multiple datasets. NegBERT detects explicit negation cues using a lexicon and infers affixal negations using WordNet, but has no mechanism for semantically implicit negation where no cue word is present.

2.4 BERT and Contextual Representation

Devlin et al. (2019) introduced BERT, a bidirectional transformer pretrained on masked language modeling and next sentence prediction. BERT can be fine-tuned with a single output layer to achieve state-of-the-art performance on a wide range of classification tasks. Luo et al. (2022) demonstrated, however, that even BERT underperforms on negation-containing samples in certain settings, noting that models risk learning surface-level distributional patterns rather than true semantic understanding. Our work investigates whether BERT's advantage on implicit negation reflects genuine semantic generalization or whether it can be replicated through targeted vocabulary augmentation of traditional models.

2.5 Positioning of This Work

While prior work has compared traditional and pretrained models in general sentiment settings, and a limited body of work has studied explicit negation, no study — to our knowledge — has (1) systematically compared traditional and pretrained models specifically on implicit negation in sentiment classification, or (2) tested whether lexicon augmentation can bridge the gap for traditional models. This paper addresses both gaps.

3 Method

3.1 Task Definition

We formulate the problem as binary sentiment classification (positive / negative). Our primary goal is not to achieve the highest possible overall accuracy, but to *diagnose* how model performance degrades across three sentence types and to assess whether targeted lexicon augmentation can close the gap to a pretrained model on implicit negation cases.

3.2 Dataset

We use the **IMDb Large Movie Review Dataset** (Maas et al., 2011), which contains 25,000 training and 25,000 test reviews with binary sentiment labels. All models are trained on the full training set and evaluated on the test set. The dataset is well-balanced (50% positive, 50% negative), so accuracy and macro F1 are closely aligned.

3.3 Negation Subset Construction

We partition the test set into three mutually exclusive subsets using keyword-based heuristic filtering:

- **Normal**: Reviews containing neither explicit nor implicit negation markers.
- **Explicit Negation**: Reviews containing one or more of the following markers: *not, no, never, nothing, nobody, nowhere, neither, nor, n't, cannot*.
- **Implicit Negation**: Reviews containing none of the above explicit markers, but containing one or more implicit negation expressions from a curated lexicon (see Section 3.4).

A single review is assigned to exactly one subset, with explicit negation taking precedence over implicit. We acknowledge that keyword-based filtering is an approximation; a review containing *hardly* could be positive in some contexts. We report the size and class balance of each subset as a quality check.

3.4 Implicit Negation Lexicon

We construct an implicit negation lexicon organized into three functional categories, each corresponding to a distinct type of implicit negation. This taxonomy is motivated by the linguistic diversity of implicit negation strategies and is used both for subset construction and for augmentation feature design.

Category A — Negative semantic verbs and adjectives: words that inherently signal negative sentiment without a negation cue, such as *avoid, disappointing, waste, regret, dread, fail, lack, forgettable, tedious, dull, unbearable, pointless, overlong, predictable, shallow, mediocre, atrocious, abysmal, laughable, insufferable*.

Category B — Result-implication constructions: multi-word phrases that imply a negative outcome through described behavior, such as *"walked out", "fell asleep", "asked for a refund", "waste of time", "waste of money"*.

Category C — Diminishing adverbials: expressions that diminish or qualify a positive quality without an explicit negation marker, such as *hardly, scarcely, barely, "far from", "anything but", "leaves much to be desired", "could have been better"*.

The three-way taxonomy reflects increasing compositional complexity: Category A cues are single tokens with stable negative polarity; Category B cues distribute the negative signal across a behavioral phrase; Category C cues require recognizing that a scalar modifier inverts an otherwise positive adjective. This structure motivates the augmentation feature design in Section 3.5.

3.5 Models

We compare five model configurations:

3.5.1 Traditional Baselines

- **Naive Bayes (NB)**: Multinomial Naive Bayes trained on TF-IDF unigram and bigram features (max_features=50,000, sublinear_tf=True).
- **Logistic Regression (LR)**: L2-regularized logistic regression (C=1.0, max_iter=1000) trained on the same TF-IDF features.

3.5.2 Augmented Traditional Models

- **NB + Lexicon**: Naive Bayes with TF-IDF features augmented with binary indicator features for each entry in the implicit negation lexicon (one feature per Category A word, one per Category B/C phrase).
- **LR + Lexicon**: Logistic Regression with the same augmented feature set.

These augmented models directly test whether explicitly encoding implicit negation vocabulary improves traditional model performance on implicit cases.

3.5.3 Pretrained Model

- **BERT**: `bert-base-uncased` fine-tuned on the IMDb training set for binary sentiment classification, using the `[CLS]` token representation with a linear classification head trained for 3 epochs with learning rate 2×10^{-5} and batch size 16.

3.6 Evaluation Metrics

We report **Accuracy** and **macro F1-score** for the full test set and each negation subset. To quantify relative degradation, we compute the **performance drop** ΔImpl. = F1(Explicit) $-$ F1(Implicit) for each model, which measures how much each model's performance deteriorates on implicit negation relative to its performance on explicit negation.

4 Results

4.1 Overall Performance

Model	Accuracy	F1
Naive Bayes	0.87256	0.87252
NB + Lexicon	0.90076	0.90076
Logistic Reg.	0.86972	0.86972
LR + Lexicon	0.89420	0.89418
BERT	0.92328	0.92328

Table 1: Overall performance on the full IMDb test set.

Table 1 shows that BERT achieves the highest overall F1 (0.923), followed by the two augmented models (NB+Lex: 0.901, LR+Lex: 0.894), and the unaugmented baselines (NB: 0.873, LR: 0.870). The consistent ordering — BERT $>$ augmented $>$ baseline — holds for both accuracy and F1, confirming that lexicon augmentation provides a meaningful overall boost but does not close the gap to contextual pretraining.

4.2 Performance by Negation Type (RQ)

	F1-score			
Model	Normal	Explicit	Implicit	ΔImpl.
NB	0.85078	0.87027	0.86233	0.00794
NB+Lex	0.84633	0.86699	0.84394	0.02305
LR	0.86791	0.90079	0.86129	0.03950
LR+Lex	0.86012	0.89321	0.88182	0.01139
BERT	0.91604	0.92127	0.92854	−0.00727

Table 2: F1-score across negation subsets. ΔImpl. = F1(Explicit) − F1(Implicit) measures performance degradation on implicit negation relative to explicit negation. Subset sizes: Normal (N=2832), Explicit (N=22113), Implicit (N=55).

The per-subset results in Table 2 reveal a clear divergence between traditional and pretrained models. BERT exhibits a *negative* performance drop (ΔImpl. = −0.007), meaning it actually performs marginally better on implicit negation than on explicit — consistent with BERT having internalized implicit sentiment associations during large-scale pretraining without needing surface cues. All traditional models, by contrast, show positive ΔImpl., with LR exhibiting the largest degradation (0.040) and NB showing a smaller but still positive drop (0.008).

The effect of lexicon augmentation is asymmetric: LR+Lexicon reduces ΔImpl. from 0.040 to 0.011 — a 72% reduction in degradation — while NB+Lexicon actually *increases* ΔImpl. from 0.008 to 0.023. This surprising reversal for NB suggests that adding new binary features to a Naive Bayes model introduces feature independence violations that slightly hurt calibration on implicit cases, even when the added features are theoretically informative. LR's L2 regularization allows it to weight new features more smoothly, enabling a genuine recovery in implicit negation performance.

After augmentation, a residual gap of approximately 4.7 F1 points separates LR+Lexicon (0.882) from BERT (0.929) on the implicit negation subset. This gap is consistent with H2 and suggests that BERT's advantage cannot be explained by vocabulary coverage alone.

Given the small size of the implicit negation subset (N=55), differences of 1–2 F1 points should be interpreted with caution. The ΔImpl. values reported here are indicative of directional trends rather than statistically precise estimates; a larger annotated benchmark would be required to draw definitive conclusions.

5 Analysis and Discussion

The results confirm H1 and partially confirm H2, with important nuances. Implicit negation is not uniformly harder than explicit negation across all models: BERT handles implicit cases at least as well as explicit ones, while traditional models show degradation of varying magnitude. The degradation is largest for LR (ΔImpl. = 0.040), consistent with LR's higher overall sensitivity to feature coverage — a missing negation feature that NB partially compensates for via other high-frequency terms creates a larger error for LR, which relies on explicit discriminative features.

Lexicon augmentation narrows the gap for LR but not NB, indicating that the utility of a curated lexicon depends on the underlying model's capacity to integrate new features gracefully. This has a practical implication: when deploying augmented traditional models, Logistic Regression is the preferred backbone, as its discriminative training objective allows it to learn reliable weights for low-frequency lexicon features from the available training signal, whereas Naive Bayes's independence assumption exacerbates the effect of artificially added binary indicators.

6 Conclusion

In this paper, we presented a systematic study of how traditional and pretrained models handle implicit negation in sentiment classification. Our contributions are: (1) a diagnostic evaluation stratified by negation type that quantifies the performance gap across model classes; and (2) a lexicon augmentation intervention that tests whether encoding implicit negation vocabulary is sufficient to close that gap.

Our results show that BERT alone avoids degradation on implicit negation while all traditional models degrade relative to explicit negation, and that lexicon augmentation selectively recovers traditional model performance for Logistic Regression (reducing ΔImpl. by 72%) but not for Naive Bayes, where the independence assumption undermines the integration of new binary features. The residual gap of approximately 4.7 F1 points separating the best augmented traditional model from BERT on the implicit negation subset suggests that contextual pretraining provides representations that a finite lexicon cannot replicate. These findings clarify when lexicon-based methods are sufficient for handling implicit negation and when contextual pretraining is necessary, with practical implications for practitioners deploying sentiment classifiers in domains where implicit negative expressions are prevalent.

7 Limitations

Our implicit negation subset is constructed using heuristic keyword matching, which may include false positives (e.g., a review containing *avoid* in a neutral context) and miss instances of implicit negation not covered by our lexicon. The subset size of 55 reviews limits the statistical power of per-subset comparisons, and results should be interpreted as indicative rather than definitive. The lexicon was constructed manually and may not generalize to other domains. Finally, our analysis is restricted to English movie reviews.

Future work could employ semantic role labeling or natural language inference to construct more precise and larger negation subsets, extend the analysis to out-of-domain datasets, and investigate whether the augmentation patterns replicate across other pretrained models such as RoBERTa or DistilBERT. A larger curated benchmark for implicit negation in sentiment analysis would substantially strengthen the conclusions drawn here.

Statement of Contributions

All authors contributed equally to the project design, implementation and report writing. Clinton focused on implementing the model configurations, running experiments, and logging results (Method and Results). Emerson worked on the per-subset performance analysis, computed ΔImpl. values, and handled formatting and tables (Introduction, Discussion, and Conclusion). Christine did the literature review and wrote the Related Work section, constructed the implicit negation lexicon, and designed the heuristic filtering rules for negative subset construction. Proofreading and edits were done collaboratively.

References

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2019)*, pages 4171–4186. Association for Computational Linguistics.

Lutfi Budi Ilmawan, Muladi, and Didik Dwi Prasetya. 2024. Negation handling for sentiment analysis task: Approaches and performance analysis. *International Journal of Electrical and Computer Engineering (IJECE)*, 14(3):3382–3393.

Aditya Khandelwal and Suraj Sawant. 2020. NegBERT: A transfer learning approach for negation detection and scope resolution. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC 2020)*, pages 5739–5748. European Language Resources Association.

Zhaoning Luo, Chenxi Li, and Dawei Zhao. 2022. Improving negation detection with negation-focused pre-training. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2022)*, pages 4188–4198. Association for Computational Linguistics.

Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. 2011. Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics (ACL 2011)*, pages 142–150. Association for Computational Linguistics.

Bo Pang, Lillian Lee, and Shivakumar Vaithyanathan. 2002. Thumbs up? Sentiment classification using machine learning techniques. In *Proceedings of the 2002 Conference on Empirical Methods in Natural Language Processing (EMNLP 2002)*, pages 79–86. Association for Computational Linguistics.

Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP 2013)*, pages 1631–1642. Association for Computational Linguistics.

Sida Wang and Christopher D. Manning. 2012. Baselines and bigrams: Simple, good sentiment and topic classification. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (ACL 2012)*, pages 90–94. Association for Computational Linguistics.